Baselines and fair comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Baselines and fair comparisons before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Hypotheses and measurable outcomes

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Replication and reproduction

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reading a research claim

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

Complete the supported Baselines and fair comparisons trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Baseline**s and fair comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Baselines and fair comparisons explanation against the supplied facts before discussing it.

Use the worked case for **Baseline**s and fair comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Model uses12 seconds and **baseline**2 on each case. Task has a3-second budget. How should results be framed?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Baselines and fair comparisons facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Baselines and fair comparisons facts and keep your reasoning.

Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Baselines and fair comparisons facts and keep your reasoning.

A report omits this qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Baselines and fair comparisons facts and keep your reasoning.

Model uses12 seconds and **baseline**2 on each case. Task has a3-second budget. How should results be framed?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied Baselines and fair comparisons facts and

> keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Baselines and fair comparisons to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **baseline**s and fair comparisons **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **baseline**s and fair comparisons.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Baselines and fair comparisons

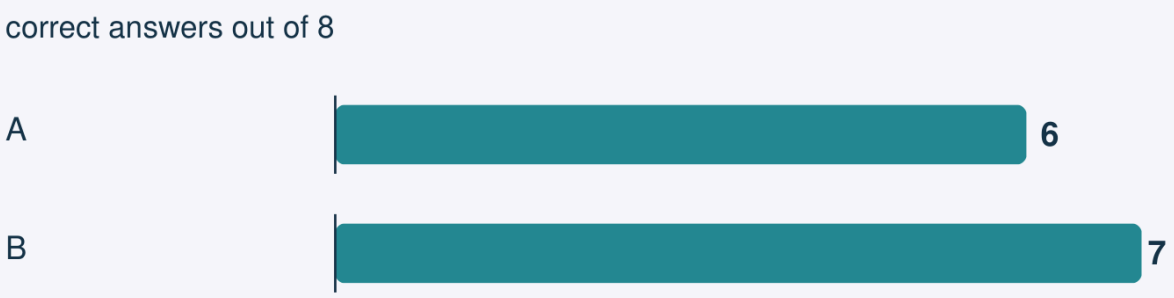
> A baseline deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources. Explain the labelled step before checking the changed condition.

A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

Simple rule A: 6/8 correct. New model B: 7/8 correct. No test case is used to tune either version.

AI ACADEMY / AI-7/8

Two versions on eight shared cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/8

Baselines and fair comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare baseline

A uses a simple documented rule rather than an intentionally weak placeholder.



2. Measure fairly

A is 75%; B is 87.5% on the same eight cases.



3. Interpret cautiously

B adds one correct case here; compare resources and failure types too.

CHANGE ONE THING

Give B the answer key before testing.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B adds one correct case here; compare resources and failure types too.

1. Declare **baseline**

2. Interpret cautiously

3. Measure fairly

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Interpret cautiously B adds one correct case here; compare resources and failure types too. A **baseline** anchors the claim of improvement.

Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give B the answer key before testing. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The resulting comparison is invalid as unseen evaluation.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The resulting comparison is

invalid as unseen evaluation. B would receive privileged information that A did not receive.

Repair a missing building block

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the **baseline** and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

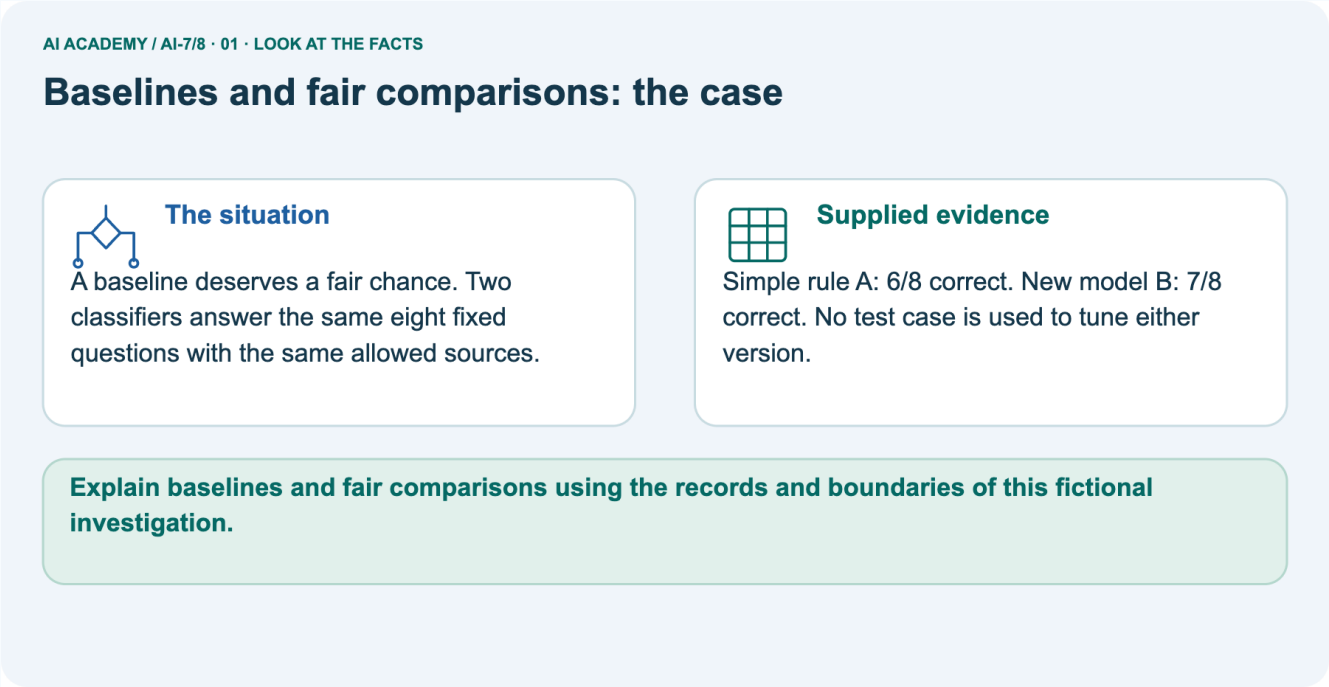
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Baselines and fair comparisons picture from its labels, then record a relevant limit.

01. Baselines and fair comparisons: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

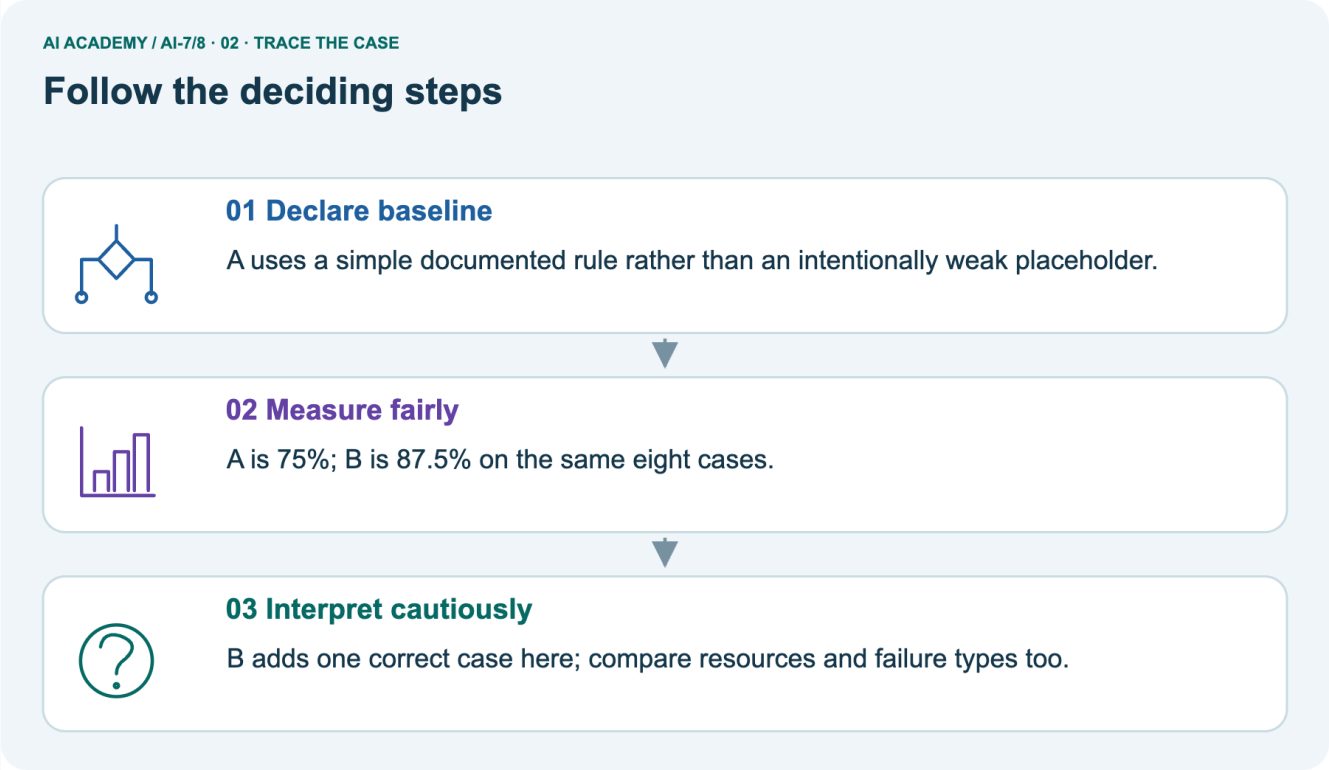
A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources. Supplied facts: Simple rule A: 6/8 correct. New model B: 7/8 correct.

No test case is used to tune either version.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



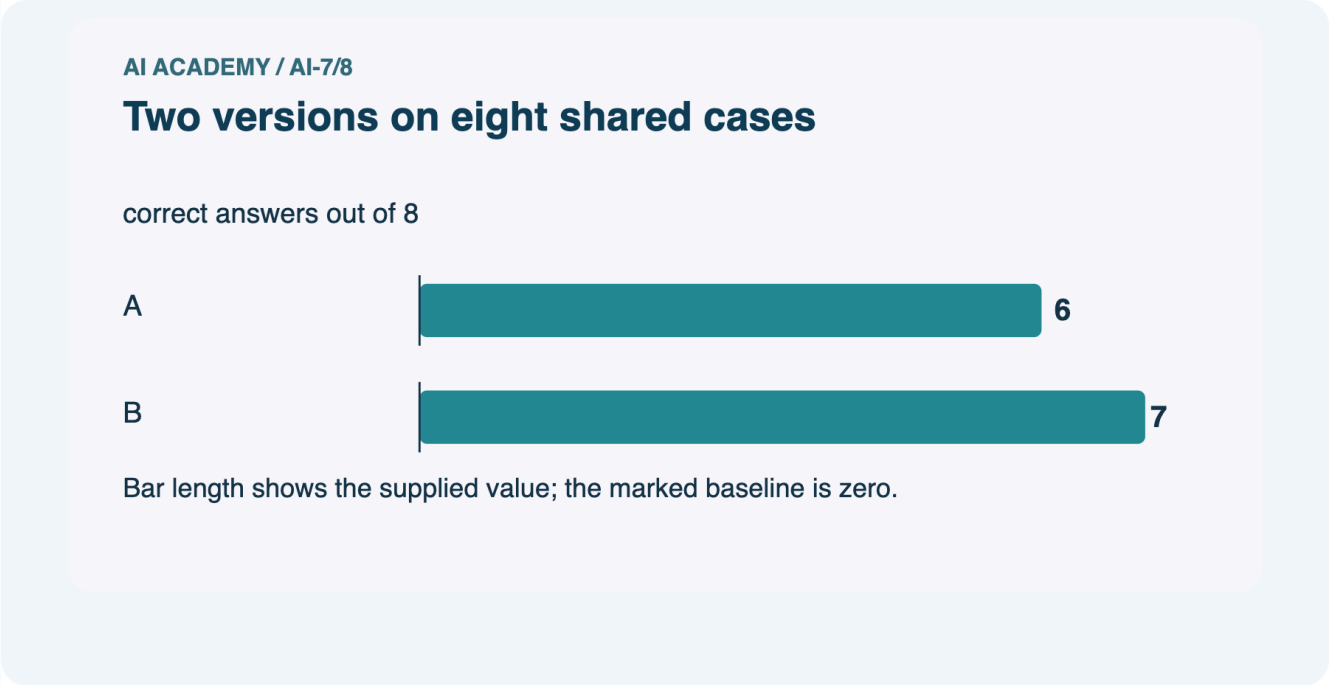
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **baseline**: A uses a simple documented rule rather than an intentionally weak placeholder. → Measure fairly: A is 75%; B is 87.5% on the same eight cases. → Interpret cautiously: B adds one correct case here; compare resources and failure types too.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two versions on eight shared cases



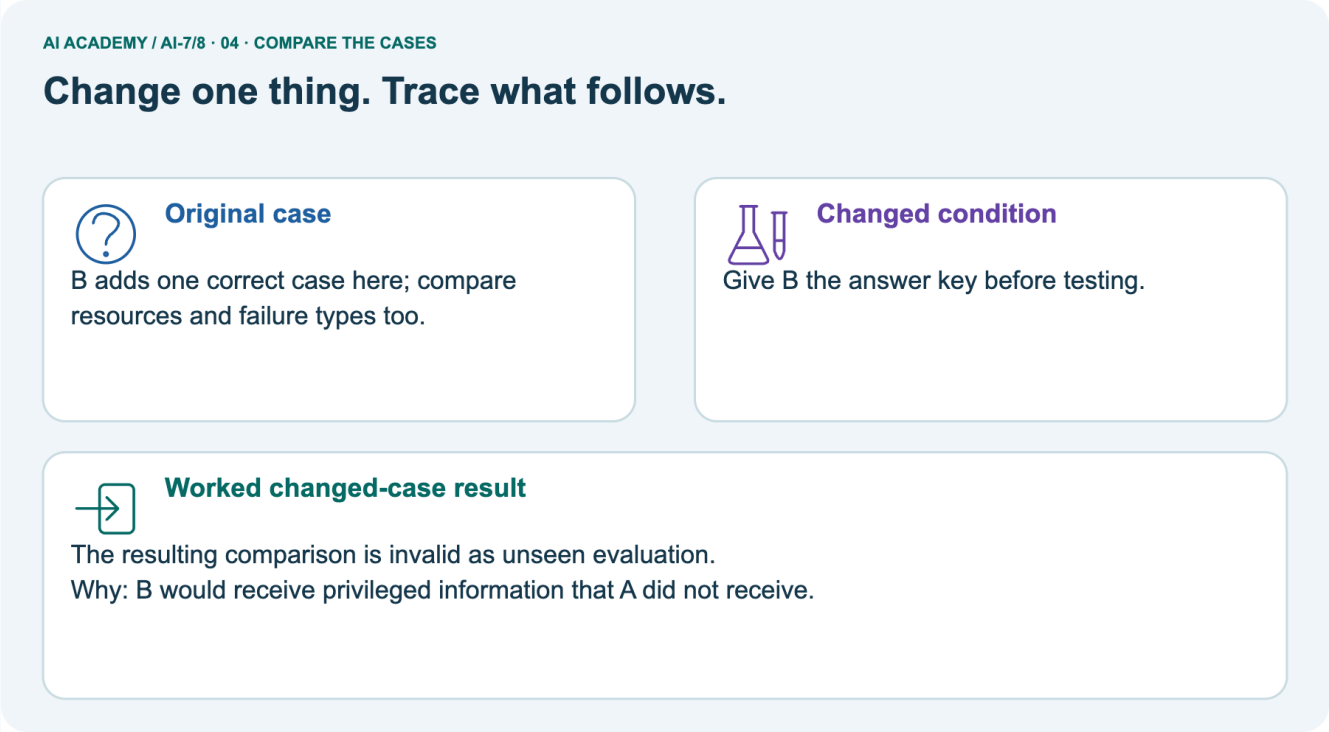
Follow the labelled picture with the matching explanation. Open the vector diagram

Two versions on eight shared cases. Units: correct answers out of 8. A: 6. B: 7.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

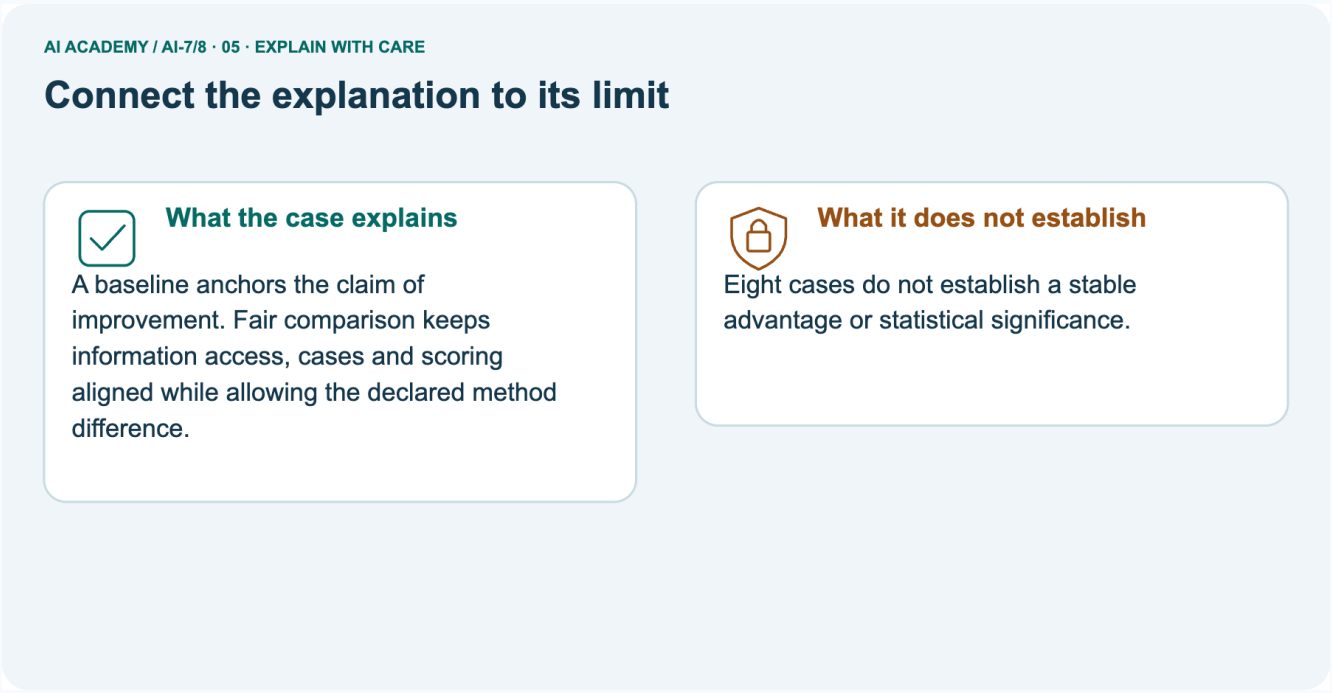
Original case: B adds one correct case here; compare resources and failure types too. Changed condition: Give B the answer key before testing. Expected result: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

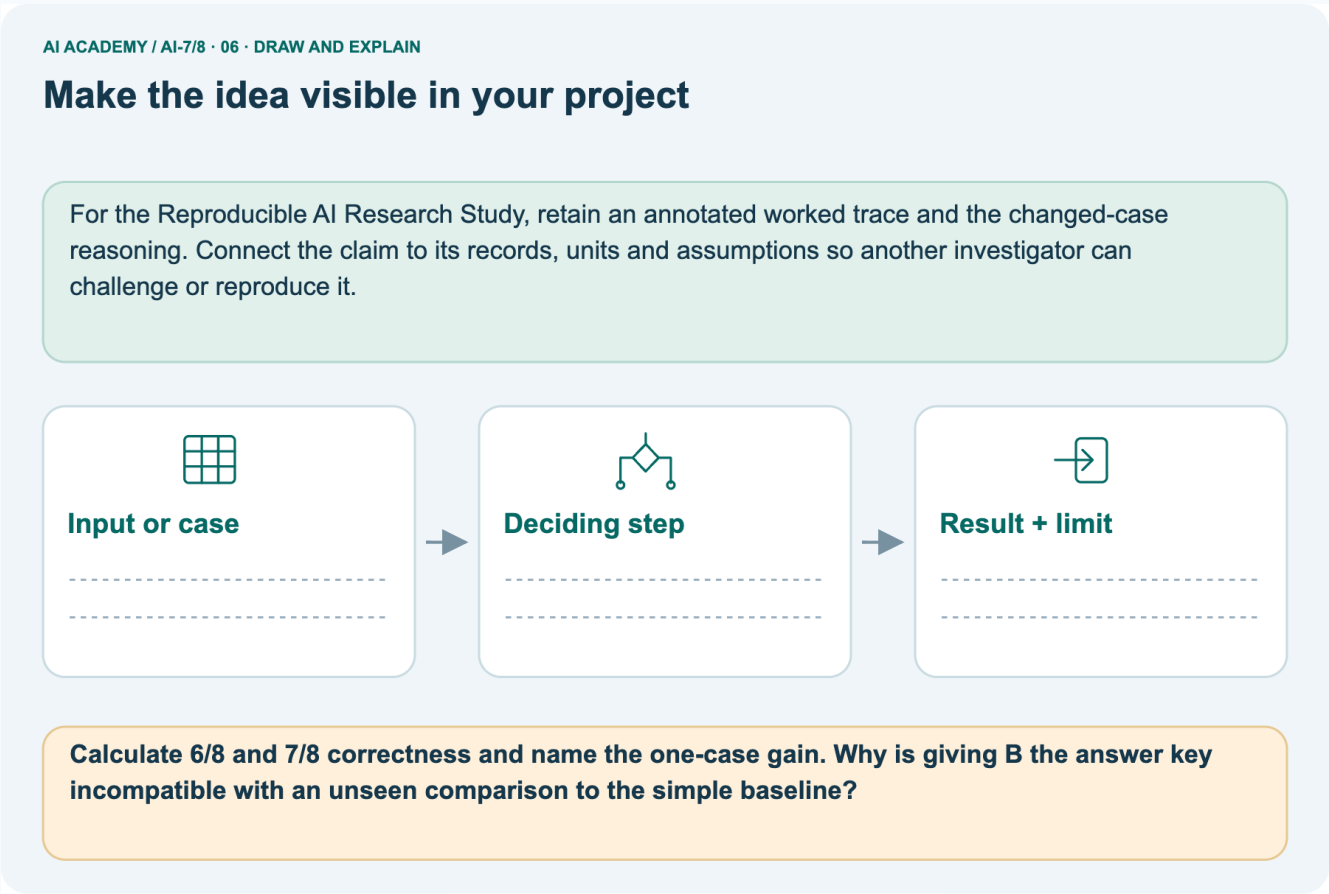
A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Limit: Eight cases do not establish a stable advantage or statistical significance.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.